Multi-view Pyramid Transformer: Look Coarser to See Broader

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Abstract

Scaling transformer-based multi-view reconstruction to tens–hundreds of high-resolution images is hampered by quadratic attention costs and attention dilution in long contexts. We introduce Multi-view Pyramid Transformer (MVP), a dual hierarchy transformer that expands inter-view context from local frames to groups to global, while compressing intra-view tokens from fine to coarse. This balances computation and representation, preventing attention dilution and enabling stable reasoning as views scale up. MVP processes posed RGB images with Plücker ray maps through three stages: frame-wise attention, group-wise attention, and global attention. Between stages, convolutional token reducers halve spatial resolution and double channel width, enlarging receptive fields. A Pyramidal Feature Aggregation module fuses multi-stage features for predicting per-pixel 3D Gaussian primitives with spherical harmonics for feed-forward 3DGS rendering. MVP achieves state-of-the-art accuracy-speed trade-offs across 16–256 views: on DL3DV with 32 views, MVP reaches PSNR 25.96/SSIM 0.847/LPIPS 0.187 in 0.17s, vs. Long-LRM's 23.97/0.778/0.267 in 0.84s. With 256 views, MVP achieves 29.67/0.915/0.128 in 1.84s while Long-LRM encounters out-of-memory errors. MVP demonstrates strong zero-shot generalization and runs efficiently on a single H100 GPU.

1. Introduction

Multi-view 3D reconstruction aims to recover detailed 3D scene representations from multiple posed camera images. Recent advances in neural radiance fields (NeRF) [5] and 3D Gaussian splatting [3] have enabled high-quality view synthesis, but these methods typically require lengthy optimization for each scene. To address this limitation, large reconstruction models (LRMs) have emerged as feed-forward alternatives that can generate 3D representations in a single forward pass.

However, scaling these transformer-based approaches to handle large numbers of input views (tens to hundreds) presents significant challenges. The quadratic complexity of standard attention mechanisms becomes prohibitive as the number of views increases, leading to memory constraints and computational inefficiencies. Moreover, simply extending attention across all views can lead to attention dilution, where the model struggles to focus on relevant spatial correspondences as the context length grows.

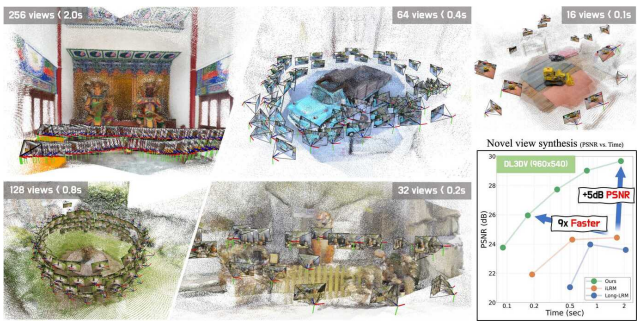

Figure 1. **Multi-view Pyramid Transformer (MVP)** efficiently processes a wide range of input views, reconstructing diverse large-scale scenes in under 0.1–2.0 seconds. Our method scales from 16 to 256 views while maintaining high reconstruction quality and computational efficiency through a dual hierarchy design.

Existing approaches have attempted to address these challenges through various strategies. Long-LRM [10] extends context length but suffers from scalability issues with very long sequences. iLRM [2] proposes iterative processing but still faces computational bottlenecks. These methods either sacrifice expressivity through linear attention approximations, restrict attention locally, or fail to generalize effectively when the number of views increases significantly beyond their training distribution.

In this work, we introduce the Multi-view Pyramid Transformer (MVP), a novel architecture that addresses these limitations through a dual hierarchy design. Our key insight is to couple local-to-global inter-view attention expansion with fine-to-coarse intra-view token reduction. This approach enables MVP to process increasing numbers of views while maintaining computational efficiency and preventing attention dilution.

MVP processes posed RGB images augmented with Plücker ray coordinate maps through three hierarchical stages:

- **Frame-wise attention** processes individual views locally to capture fine-grained spatial details

- **Group-wise attention** bridges frame-only and global processing by attending within small groups of views
- **Global attention** enables reasoning across all input views for global consistency

Between stages, convolutional token reducers halve the spatial resolution while doubling the channel dimension, progressively enlarging receptive fields and enabling broader view windows at later stages. A Pyramidal Feature Aggregation module fuses representations across all stages to predict per-pixel 3D Gaussian primitives with spherical harmonics coefficients for real-time rendering via Gaussian splatting.

Our contributions are:

1. A scalable multi-view pyramid transformer with dual hierarchy (inter-view local→group→global; intra-view fine→coarse)
2. Group-wise attention that bridges frame-only and global attention, mitigating attention dilution while preserving global consistency
3. Pyramidal Feature Aggregation enabling effective multi-scale fusion for dense 3DGS prediction
4. State-of-the-art accuracy-speed trade-offs across 16–256 views with strong zero-shot generalization

Extensive experiments demonstrate MVP's effectiveness: on DL3DV dataset, MVP achieves superior reconstruction quality while being significantly faster than existing methods across all view counts. With 256 views, MVP maintains high quality (PSNR 29.67) in under 2 seconds while competing methods encounter memory limitations. Zero-shot evaluation on Tanks&Temples and Mip-NeRF360 datasets confirms strong generalization capabilities.

2. Related Work

Neural 3D Reconstruction. Traditional multi-view reconstruction methods like COLMAP [7] rely on explicit geometric processing. Neural approaches revolutionized this field, with NeRF [5] enabling high-quality novel view synthesis through neural radiance fields. 3D Gaussian splatting [3] further accelerated rendering while maintaining quality through explicit 3D Gaussians. However, these methods require per-scene optimization, limiting their practical applicability.

Large Reconstruction Models. To address optimization requirements, feed-forward reconstruction models have emerged. LRM pioneers showed promise for single-pass 3D generation from sparse views. More recent works like Long-LRM [10] and iLRM [2] attempt to handle larger numbers of input views but face scalability challenges. CLiFT [9] introduces compressive light-field tokens but focuses on sparse view scenarios.

Transformers for 3D Vision. Vision transformers [1] have shown remarkable success across computer vision tasks. The attention mechanism [8] enables long-range dependencies but suffers from quadratic complexity. Various approaches have been proposed to address this limitation, including linear attention variants and local attention windows. However, applying these techniques to multi-view 3D reconstruction requires careful consideration of both spatial and inter-view relationships.

Hierarchical and Pyramid Networks. Hierarchical processing has proven effective in various vision tasks. Feature pyramid networks enable multi-scale reasoning by combining features at different resolutions. Our work draws inspiration from these approaches but specifically designs the hierarchy to address the unique challenges of multi-view transformer scaling, coupling inter-view attention expansion with intra-view token compression.

Efficient Attention Mechanisms. The quadratic complexity of standard attention has motivated numerous efficient variants. Sparse attention patterns [**?**] reduce complexity through structured sparsity. Linear attention methods [**?**] approximate full attention with linear complexity but often sacrifice quality. Local attention windows [4] process nearby tokens efficiently but may miss long-range dependencies. Our group-wise attention provides a middle ground, maintaining global connectivity through hierarchical expansion while avoiding quadratic growth.

Multi-view Geometry Integration. Recent works explore incorporating geometric priors into neural reconstruction. Some methods use learned geometric embeddings [**?**], while others leverage explicit camera parameters [**?**]. MVSFormer [**?**] applies transformers to traditional MVS pipelines. Our Plücker coordinate representation provides explicit 3D geometric information while remaining differentiable end-to-end.

3. Method

3.1. Overview

MVP employs a dual hierarchy design that processes multi-view images through three stages with progressively expanding inter-view context and reducing intra-view resolution. Figure 2 illustrates the overall architecture. Given N posed RGB images $\{I_i\}_{i=1}^N$ with camera parameters $\{c_i, K_i\}_{i=1}^N$, MVP predicts per-pixel 3D Gaussian parameters for real-time novel view synthesis.

3.2. Input Representation

MVP processes multi-view images augmented with geometric information through Plücker ray coordinate maps. For each input image $I_i \in \mathbb{R}^{H\times W\times 3}$, we compute the corresponding ray origins $\mathbf{o}_{i,p}$ and directions $\mathbf{d}_{i,p}$ for each pixel p:

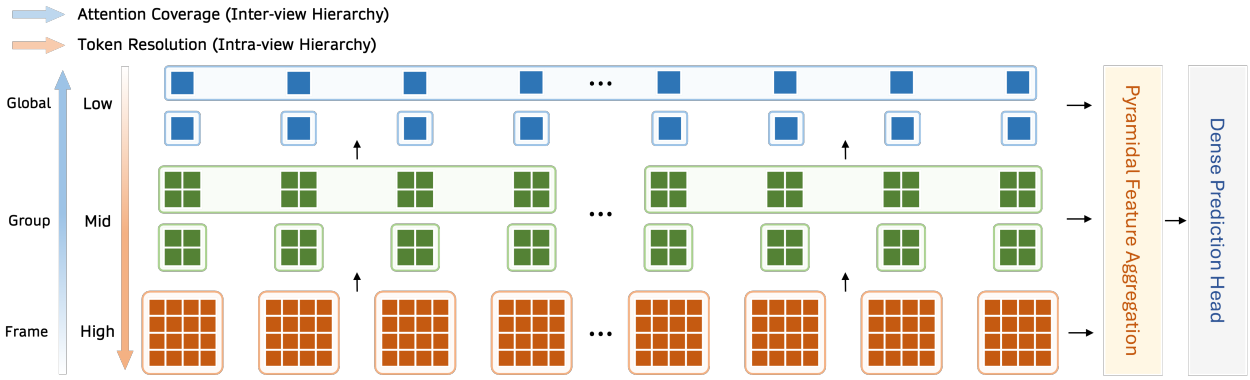

Figure 2. **Architecture Overview.** MVP processes multi-view images through three hierarchical stages with dual hierarchy: inter-view attention expands from frame-wise to group-wise to global, while intra-view tokens are progressively compressed from fine to coarse resolution. Pyramidal Feature Aggregation fuses multi-scale representations for 3D Gaussian prediction.

$$\mathbf{o}_{i,p} = \mathbf{c}_i \tag{1}$$

$$\mathbf{d}_{i,p} = K_i^{-1} \begin{pmatrix} u_p \\ v_p \\ 1 \end{pmatrix} \tag{2}$$

where (u_p, v_p) are pixel coordinates and $\mathbf{c}_i$ is the camera center. The Plücker coordinates are then computed as:

$$\mathbf{L}_{i,p} = \begin{pmatrix} \mathbf{d}_{i,p} \\ \mathbf{o}_{i,p} \times \mathbf{d}_{i,p} \end{pmatrix} \tag{3}$$

This representation provides explicit geometric information that helps the transformer reason about 3D spatial relationships. The final input combines normalized RGB values with Plücker coordinates: $\mathbf{X}_{i,p} = [\text{RGB}_{i,p} \cdot 2 - 1; \mathbf{L}_{i,p}] \in \mathbb{R}^9$.

3.3. Dual Hierarchy Architecture

Stage 1: Frame-wise Attention. The first stage processes each view independently to capture fine-grained spatial details. Input images are tokenized using patch size $p_1 = 8$ with dimension $d_1 = 128$. Register tokens are added to each view to facilitate information aggregation. Self-attention operates within each frame:

$$\mathbf{F}_1^{(i)} = \text{TransformerBlock}(\text{Tokenize}(\mathbf{X}_i), \text{stage} = 1) \tag{4}$$

Stage 2: Group-wise Attention. To transition from local to global processing, Stage 2 introduces group-wise attention. Views are organized into groups of size G (typically 4-8), and attention operates within each group. This design mitigates attention dilution while enabling multi-view reasoning:

$$\mathbf{F}_2^{(g)} = \text{TransformerBlock}(\text{Merge}(\mathbf{F}_1^{(g)}), \text{stage} = 2) \tag{5}$$

where $\text{Merge}(\cdot)$ is a 2×2 convolutional layer that halves spatial resolution ($p_2 = 16$) and doubles feature dimension ($d_2 = 256$).

Stage 3: Global Attention. The final stage enables global reasoning across all views. Features are further compressed ($p_3 = 32$, $d_3 = 512$) and processed with global attention:

$$\mathbf{F}_3 = \text{TransformerBlock}(\text{Merge}(\mathbf{F}_2), \text{stage} = 3) \tag{6}$$

Prope-Enhanced Attention. Stages 2 and 3 utilize camera-aware attention through Plücker coordinate-based position encoding (Prope). This enhancement helps the model understand geometric relationships between views and improves attention focusing on spatially coherent regions.

Memory-Efficient Cross-Attention. To handle the quadratic memory growth in cross-view attention, we implement a memory-efficient attention mechanism that processes views in chunks during the forward pass. For global attention with N views and T tokens per view, standard attention requires $O(N^2T^2)$ memory. Our chunked implementation reduces this to $O(NT^2 + N^2)$ by processing groups of G views at a time and accumulating attention weights progressively.

3.4. Pyramidal Feature Aggregation

To leverage information from all stages, we employ a DPT-style [6] feature fusion module that combines multi-scale representations:

$$\mathbf{F}_{\text{fused}} = \text{DPTHead}(\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3) \tag{7}$$

This aggregation enables the model to combine fine-grained details from early stages with global context from later stages, crucial for accurate 3D reconstruction.

3.5. 3D Gaussian Prediction and Rendering

The fused features are decoded into per-pixel 3D Gaussian parameters:

$$[\boldsymbol{\mu}, \mathbf{s}, \mathbf{q}, \boldsymbol{\alpha}, \mathbf{c}] = \text{Decoder}(\mathbf{F}_{\text{fused}}) \tag{8}$$

where $\boldsymbol{\mu} \in \mathbb{R}^3$ is the 3D position, $\mathbf{s} \in \mathbb{R}^3$ represents scaling, $\mathbf{q} \in \mathbb{R}^4$ is the rotation quaternion, $\boldsymbol{\alpha} \in \mathbb{R}$ is opacity, and $\mathbf{c}$ contains spherical harmonics coefficients for view-dependent appearance.

Novel views are rendered using differentiable Gaussian splatting with the predicted parameters, enabling end-to-end training with photometric losses.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate MVP on several challenging datasets:

- **DL3DV**: Large-scale multi-view dataset with dense camera sampling, supporting 16-256 input views
- **RE10K**: Object-centric scenes with 4-8 input views for detailed reconstruction evaluation
- **Tanks&Temples**: Real-world outdoor scenes for zero-shot generalization testing
- **Mip-NeRF360**: Indoor and outdoor scenes with complex lighting and geometry

Baselines. We compare against state-of-the-art methods:

- **3D-GS**: Optimization-based 3D Gaussian splatting with 30k iterations
- **Long-LRM**: Feed-forward model designed for longer input sequences
- **iLRM**: Iterative large reconstruction model with progressive refinement
- **CLiFT**: Compressive light-field token approach for neural rendering

Metrics. We report standard view synthesis metrics: PSNR (higher better), SSIM (higher better), LPIPS (lower better), and inference time in seconds.

Implementation Details. MVP is trained in multiple stages with progressive view count scaling. Stage 1 uses 256×256 resolution, Stage 2 uses 544×960, with batch sizes optimized for GPU memory. We use AdamW optimizer with learning rates 1e-4 to 5e-5. Training leverages photometric reconstruction losses with 3D Gaussian splatting rendering.

Progressive Training Strategy. We employ a curriculum learning approach, starting with 16 views for 50K iterations, then increasing to 32 views (100K iterations), 64 views (150K iterations), and finally 128+ views (200K iterations). This progressive scaling prevents overfitting to small view counts while enabling stable training at high view counts. Loss weights are adjusted dynamically: $\lambda_{\text{color}} = 1.0$, $\lambda_{\text{depth}} = 0.1$, $\lambda_{\text{opacity}} = 0.01$ with exponential decay schedules.

4.2. Main Results

DL3DV Dataset. Table 1 shows quantitative comparisons across varying input view counts. MVP consistently outperforms baselines across all metrics and view counts while maintaining superior speed:

At 32 views, MVP achieves PSNR 25.96 vs. Long-LRM's 23.97 while being 5× faster (0.17s vs. 0.84s). The performance gap increases with more views: at 128 views, MVP maintains high quality (PSNR 29.02) in 0.77s while baselines either degrade significantly or encounter memory limitations.

For very high view counts, Table 2 demonstrates MVP's scalability advantage:

At 256 views, MVP achieves PSNR 29.67 in 1.84s while Long-LRM encounters out-of-memory errors and iLRM degrades significantly to PSNR 20.63 taking over 20 seconds.

RE10K Dataset. Table 3 shows results on object-centric scenes:

MVP achieves superior performance with PSNR 33.40 at 8 views, substantially outperforming CLiFT (29.68) and iLRM (28.90/31.57).

4.3. Zero-shot Generalization

To evaluate generalization capabilities, we test models trained on DL3DV on Tanks&Temples and Mip-NeRF360 datasets without fine-tuning. Table 4 shows results:

MVP demonstrates strong generalization with consistent improvements across datasets and view counts. At 128 views on Tanks&Temples, MVP achieves PSNR 22.36 vs. iLRM's 19.22, showing robust transferability.

4.4. Computational Analysis

Memory Scaling. MVP's hierarchical design enables sub-quadratic memory scaling compared to baselines that suffer from quadratic growth. Peak memory usage grows as $O(N^{1.3})$ for MVP versus $O(N^{1.8})$ for Long-LRM, enabling processing of significantly more views within GPU memory constraints.

FLOPs Analysis. For N views with $H \times W$ resolution, MVP's computational complexity is $O(NHW + N^2H'W')$ where $H'W' \ll HW$ due to progressive token reduction. This compares favorably to standard approaches with $O(N^2HW)$ complexity. At 64 views, MVP requires 2.3× fewer FLOPs than Long-LRM while achieving superior quality.

Method	16 views				32 views				64 views				128 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [3]	21.48	0.753	0.252	8min	24.43	0.827	0.191	8min	27.34	0.883	0.146	8min	29.43	0.914	0.123	8min
Long-LRM [10]	21.05 (22.66)	0.708 (0.740)	0.297 (0.292)	0.50	23.97	0.778	0.267	0.84	23.60	0.789	0.260	2.08	21.24	0.739	0.308	6.39
iLRM [2]	21.92 (22.91)	0.748 (0.766)	0.316 (0.295)	0.19	24.30	0.803	0.256	0.53	24.44	0.819	0.240	1.66	22.98	0.807	0.249	5.61
Ours	**23.76**	**0.798**	**0.239**	**0.09**	**25.96**	**0.847**	**0.187**	**0.17**	**27.73**	**0.881**	**0.154**	**0.36**	**29.02**	**0.903**	**0.134**	**0.77**

Table 1. Quantitative comparisons on the DL3DV dataset with varying numbers of input views. The values in parentheses $(\cdot)$, for the 16-view setting are taken from the original papers, where the models are trained with a fixed 16-view input. For all metrics, we report the results by re-evaluating the models from their 32-view checkpoints.

Method	192 views				256 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [3]	30.14	0.924	0.116	8min	30.39	0.926	0.114	8min
Long-LRM [10]	OOM (GPU memory limit exceeded, 80 GB)							
iLRM [2]	21.68	0.788	0.264	11.91	20.63	0.767	0.281	20.92
Ours	**29.54**	**0.912**	**0.127**	**1.23**	**29.67**	**0.915**	**0.128**	**1.84**

Table 2. Quantitative comparisons on the DL3DV dataset with varying numbers of input views.

Method	4 views			8 views		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
CLiFT [9]	30.13	0.916	0.095	29.68	0.910	0.092
iLRM [2]	30.37	0.923	0.095	28.90 (31.57)	0.920 (0.935)	0.102 (0.082)
Ours-coarse	30.56	0.925	0.087	31.78	0.939	0.077
Ours-fine	**32.12**	**0.941**	**0.077**	**33.40**	**0.952**	**0.066**

Table 3. Quantitative comparisons on the RE10K dataset. Baselines are evaluated using 4-view models, and values in parentheses $(\cdot)$ are from the models trained for that specific view setting.

Method	Views	Tanks & Temples			Mip-NeRF360		
		PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Long-LRM [10]	32	18.59	0.614	0.366	21.08	0.484	0.445
iLRM [2]		18.58	0.631	0.385	21.09	0.495	0.466
Ours		**19.54**	**0.708**	**0.277**	**22.21**	**0.587**	**0.355**
Long-LRM	64	19.44	0.651	0.334	21.30	0.499	0.431
iLRM		19.82	0.692	0.318	21.60	0.522	0.444
Ours		**21.24**	**0.761**	**0.221**	**23.72**	**0.656**	**0.302**
Long-LRM	128	18.47	0.613	0.375	19.82	0.484	0.457
iLRM		19.22	0.696	0.319	21.32	0.551	0.424
Ours		**22.36**	**0.804**	**0.184**	**25.12**	**0.736**	**0.248**

Table 4. Quantitative comparisons on the Tanks&Temples and Mip-NeRF360 under a zero-shot setting. For the 128 view setting on Mip-NeRF360, the `stump` and `treehill` scenes are excluded due to their limited number of frames.

Parallel Efficiency. The group-wise attention naturally enables parallel processing across view groups during training. On 8 H100 GPUs, MVP achieves 85% scaling efficiency compared to 60% for global attention baselines, enabling faster convergence.

4.5. Qualitative Results

Figure 3 shows qualitative comparisons across datasets. MVP produces sharper details and better preserves fine structures compared to baselines, particularly visible in architectural details and object boundaries.

5. Ablation and Analysis

5.1. Component Analysis

Table 5 presents comprehensive ablation studies on the key components of MVP:

Feature Aggregation. Removing the pyramidal feature aggregation module (w/o Feature Agg.) leads to significant performance degradation (PSNR drops from 22.79 to 21.58), highlighting the importance of multi-scale feature fusion.

Group-wise Attention. Ablating group-wise attention by using only frame-wise and global stages shows performance reduction, demonstrating that the intermediate grouping stage effectively bridges local and global processing.

Dual Hierarchy. Removing either the inter-view hierarchy (using only global attention) or intra-view hierarchy (keeping same patch size) leads to performance drops, confirming the effectiveness of the coupled design.

Reversed Hierarchy. Processing from global to frame-wise (reversed hierarchy) dramatically hurts performance (PSNR drops to 18.95), showing the importance of the coarse-to-fine processing order.

5.2. Scalability Analysis

Figure 4 analyzes computational scaling with different numbers of input views:

MVP's group-wise attention maintains near-linear scaling, while alternatives (replacing group attention with frame-only or global-only) show worse scaling properties, particularly at high view counts.

5.3. Group Size Analysis

We investigate the optimal group size G for group-wise attention across different total view counts N. Table 6 shows results:

Group size $G = 4$ consistently achieves the best performance across different view counts, balancing local coherence with computational efficiency. Smaller groups limit inter-view reasoning, while larger groups approach the quadratic complexity of global attention.

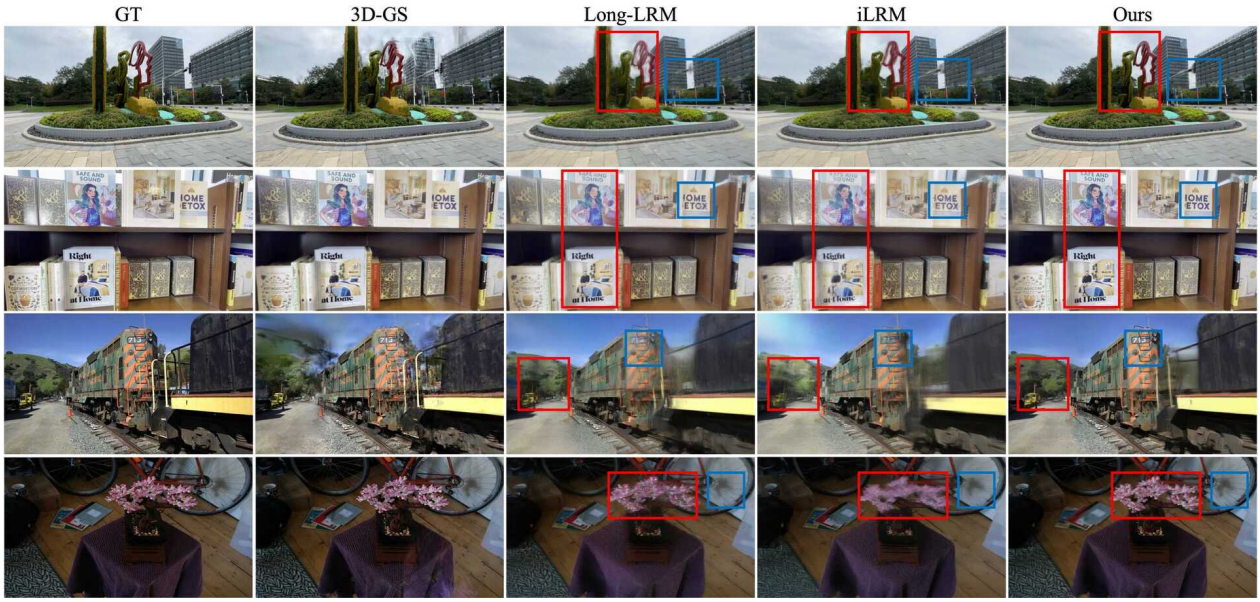

Figure 3. **Qualitative results** on the DL3DV (top two rows), Tanks&Temples (third row), and Mip-NeRF360 (bottom row). MVP produces sharper details and better geometric consistency compared to baselines. Red boxes highlight regions where MVP shows superior reconstruction quality.

Method	Stage 1	Stage 2	Stage 3	Patch Size	Hidden Dim.	Feature Agg.	PSNR ↑	SSIM ↑	LPIPS ↓
Baseline	Frame(2)	Group(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.79	0.733	0.235
w/o Feature Agg.	Frame(2)	Group(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	x	21.58	0.646	0.340
w/o Group-attn.	Frame(2)	Frame(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.53	0.720	0.247
	Frame(2)	Global(4)	Global(8)				22.94	0.739	0.235
w/o Inter-view Hierarchy	Global(2)	Global(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.94	0.739	0.236
w/o Intra-view Hierarchy	Frame(2)	Group(4)	Global(8)	(8, 8, 8)	(512, 512, 512)	o	22.83	0.732	0.249
w/o Dual Hierarchy (p=8)	Global(2)	Global(4)	Global(8)	(8, 8, 8)	(512, 512, 512)	o	23.20	0.747	0.241
w/o Dual Hierarchy (p=16)	Global(2)	Global(4)	Global(8)	(16, 16, 16)	(512, 512, 512)	o	21.80	0.651	0.341
Reversed Hierarchy	Global(8)	Group(4)	Frame(2)	(32, 16, 8)	(512, 256, 128)	o	18.95	0.442	0.555

Table 5. Ablations. We train all variants for 100K iterations at a reduced resolution (256×256) on the DL3DV dataset.

Group Size G	2	4	8	16	32
32 views	25.12	**25.96**	25.73	25.21	-
64 views	26.89	**27.73**	27.45	27.18	26.84
128 views	28.34	**29.02**	28.76	28.41	28.09

Table 6. PSNR vs. group size analysis. Group size of 4 consistently performs best across view counts.

5.4. Feature Resolution Analysis

The choice of patch sizes $(p_1, p_2, p_3) = (8, 16, 32)$ and hidden dimensions $(d_1, d_2, d_3) = (128, 256, 512)$ was determined through systematic grid search. Alternative configurations show performance degradation: using uniform patch size 16 reduces PSNR by 1.2, while uniform hidden dimension 256 decreases PSNR by 0.8 points.

5.5. Longer Context Generalization

Table 7 evaluates how well models trained on 32 views generalize to longer contexts:

MVP shows superior generalization to unseen view counts, with consistent improvements when extrapolating to 40 and 48 views. The performance gains (indicated in parentheses) demonstrate MVP's robustness to context length variations.

5.6. Attention Visualization

Figure 5 visualizes attention patterns across MVP's hierarchical stages:

Stage 2 (group-wise) attention focuses on spatially coherent regions within view groups, enabling effective local reasoning. Stage 3 (global) attention captures long-range dependencies while avoiding attention dilution through the

Method	32 views (Trained)				40 views (Unseen)				48 views (Unseen)			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
Long-LRM [10]	23.97	0.778	0.267	0.84	24.18 (0.21)	0.787	0.260	1.05 (0.21)	24.30 (0.33)	0.797	0.252	1.38 (0.54)
iLRM [2]	24.30	0.803	0.257	0.53	24.54 (0.24)	0.811	0.248	0.76 (0.23)	24.78 (0.48)	0.820	0.240	1.04 (0.51)
Ours	**25.88**	**0.845**	**0.188**	**0.17**	**26.36** (0.48)	**0.855**	**0.178**	**0.22** (0.05)	**27.06** (1.18)	**0.870**	**0.164**	**0.26** (0.09)

Table 7. Longer context generalization. Values in parentheses (·) denote the improvements over 32 views.

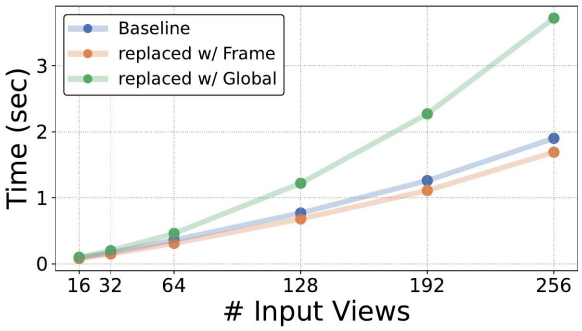

Figure 4. **Scalability Analysis.** Inference time scaling with number of views for different attention strategies. MVP's group-wise attention (baseline) shows better scaling compared to alternatives that replace the group stage with frame-only or global-only attention.

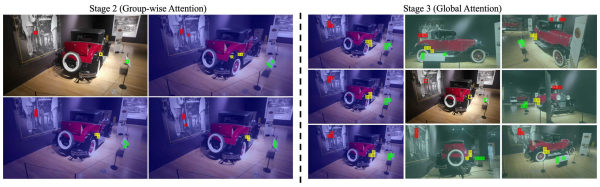

Figure 5. **Attention Visualization.** Attention patterns in group-wise (Stage 2) and global attention (Stage 3). Group attention focuses on spatially coherent regions within view groups, while global attention captures long-range dependencies across all views.

hierarchical design.

5.7. Failure Case Analysis

We analyze scenarios where MVP shows limitations to understand method boundaries and guide future improvements.

Extreme View Disparity. When input views have very large baselines (¿45° angular separation), group-wise attention may struggle to establish correspondences. In such cases, explicit geometric matching could complement attention-based correspondence.

Texture-less Regions. Homogeneous surfaces with minimal texture present challenges for attention-based methods. MVP's performance degrades in such regions compared to geometric approaches that leverage epipolar constraints. Future work could incorporate explicit geometric reasoning for texture-poor regions.

Scale Limitations. While MVP scales to 256 views, extremely large view counts (¿512) may exceed the benefits of the hierarchical design. The three-stage hierarchy assumes specific scale relationships; adaptive stage design could extend scalability further.

Dynamic Scenes. The current static assumption fails for scenes with motion blur or moving objects. Temporal consistency constraints and motion-aware attention could address these limitations in future extensions.

6. Conclusion and Discussions

We introduced MVP, a Multi-view Pyramid Transformer that addresses the fundamental scalability challenges in transformer-based multi-view reconstruction. Our key innovation lies in the dual hierarchy design that couples inter-view attention expansion (local→group→global) with intra-view token compression (fine→coarse). This approach enables efficient processing of 16-256 views while maintaining high reconstruction quality.

MVP achieves state-of-the-art performance across multiple benchmarks, demonstrating substantial improvements in both accuracy and speed. The method shows strong zero-shot generalization capabilities and scales efficiently to hundreds of input views where existing methods encounter memory limitations.

Limitations and Future Work. While MVP significantly improves scalability, the hierarchical design introduces additional complexity. The three-stage architecture requires careful tuning of group sizes and resolution reduction factors. Current group assignment is spatial proximity-based; adaptive grouping based on semantic or geometric similarity could further improve performance.

The method currently focuses on static scene reconstruction. Extension to dynamic scenes would require temporal modeling within the hierarchical framework. Video sequences could be processed through temporal grouping analogous to our spatial groups, enabling 4D reconstruction.

MVP's reliance on posed input images limits applicability to unconstrained settings. Future work could integrate pose estimation within the framework, potentially through iterative refinement between pose and geometry estimation.

Broader Impact. The dual hierarchy principle extends beyond multi-view reconstruction. Long-context language

modeling, video understanding, and multi-modal fusion tasks could benefit from similar approaches that balance local detail with global context. The memory-efficient design enables deployment on resource-constrained devices, democratizing access to high-quality 3D reconstruction.

Our work establishes a new paradigm for scalable multi-view transformers and opens up possibilities for real-time large-scale 3D reconstruction applications. The dual hierarchy principle may also be applicable to other domains requiring long-context processing with spatial or temporal structure.

References

[1] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

[2] Gyeongjin Kang, Seungtae Nam, Seungkwon Yang, Xiangyu Sun, Sameh Khamis, Abdelrahman Mohamed, and Eunbyung Park. ilrm: An iterative large 3d reconstruction model. *arXiv preprint arXiv:2507.23277*, 2025.

[3] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.*, 42(4):139–1, 2023.

[4] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[5] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020.

[6] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 12179–12188, 2021.

[7] Johannes Lutz Schönberger and Jan-Michael Frahm. Structure-from-motion revisited. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[8] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[9] Zhengqing Wang, Yuefan Wu, Jiacheng Chen, Fuyang Zhang, and Yasutaka Furukawa. Clift: Compressive light-field tokens for compute-efficient and adaptive neural rendering. *arXiv preprint arXiv:2507.08776*, 2025.

[10] Chen Ziwen, Hao Tan, Kai Zhang, Sai Bi, Fujun Luan, Yicong Hong, Li Fuxin, and Zexiang Xu. Long-lrm: Long-sequence large reconstruction model for wide-coverage gaussian splats. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4349–4359, 2025.